

Vulnerable Service

- Metasploitable2- FTP server
- Vulnerable
- tleemc2/metasploitable2 and vsftpd:latest
- 21/tcp
- FTP also known as file transfer protocol provides file listing, download, and upload files over a tcp port. The main vulnerability it has in Metasploitable is the backdoor it provides which is when an attacker logs in with a specific character in the name “:” the backdoor will make a remote shell which gives remote command shell access to an attacker. I decided to make this known for Red team because I believe that it is an easy vulnerability to get but can be a bit annoying to understand fully so it can also make it seem like it is too good to be true.

Decoy Service 1

- NGINX TLS-backend web server
- Decoy
- nginx:latest
- 80/tcp
- NGINX is a webserver that serves HTTPS content on its port. It is a good decoy because the web server front ends can be found everywhere in production and the docker image is well-maintained. It will attract for broad scanning and to an attacker it will look like a real site with a valid TLS cert and a normal server banner.

Decoy Service 2

- PostgreSQL
- Decoy
- Postgres:latest
- 5432/tcp
- PostgreSQL is a relational database that is used mostly in production backends, making it attractive for attackers. The current image deployment can be locked down. Attackers may detect that it is an actual real DB but have no way to get the actual data from the database. To an attacker it should look like an actual database with data inside of it as well with a schema. But unknown queries to the DB will be detectable and easily shut down.

Network Architecture Design:

- NETWORKPAULO
- Metasploitable2 FTP: 21/tcp --> 2121/tcp
- Decoy 1 Nginx: 80/tcp → 2431/tcp
- Decoy 2 PostgreSQL: 5432/tcp -> 3333/tcp

